

Elijah Zhao

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EDUCATION

University of British Columbia

Bachelor of Science, Major in Computer Science, Minor in Math

GPA: 4.1/4.33

Expected Graduation: May 2028

EXPERIENCE

Hootsuite

May 2026 – Dec 2026

Software Engineer Intern, Platform Infrastructure

Vancouver, BC

- Built a **Java Apache Calcite**-based **JDBC** connector for Hootsuite Analytics, exposing data as relational tables and using filter pushdown to optimize Tableau SQL queries; handled **5,000+** queries in its first week.
- Built the **first CI-gated Playwright E2E suite** for an MCP connector platform spanning three services; mocked flaky MCP and OAuth dependencies in a Docker Compose environment and covered **10+** failure paths.
- Tracked Analytics API usage with Go middleware, streaming request events through **Kafka** to Mixpanel.
- Built App Directory features in **Go**, including PKCE OAuth and APIs for MCP connector registration and discovery.

Nexera Robotics (acquired by Locus Robotics)

January 2026 – April 2026

Software Engineer Intern

Vancouver, BC

- Led redesign of a robotic pick-cycle pipeline in Python/C++, parallelizing ML inference and NumPy-based decision logic with **Ray Core** and optimizing critical paths in C++, improving throughput by **30% (7 to 10 picks/min)**.
- Built internal robotics tooling in Python/PostgreSQL interfacing with **Universal Robots** firmware, shipping **5+** features including custom speed/pathing controls and performance tracking to identify system bottlenecks.
- Developed a dataset curation pipeline that reduced **1M+** unfiltered images to **7K** using **DINOv2** embeddings and **pgvector** similarity search for deduplication and sampling; stored images in S3 and metadata/vectors in RDS.

UBC BizTech

August 2025 – Present

Software Development Lead

Vancouver, BC

- Leading **9** developers in building software and robotics platforms for UBC's largest tech club, serving **2,000+** users.
- Built a serverless backend with **AWS Lambda + Cognito** to manage authentication, payments, and event workflows; integrated **Stripe** webhooks to process **\$30K+** in membership payments and event confirmations.
- Designed **DynamoDB** access patterns with **GSIs** enabling queries across **9+** events and **900+** members.
- Shipped custom event products, including a membership portal, NFC networking system, and mock stock exchange.

University of British Columbia

January 2026 – April 2026

Teaching Assistant, Data Structures & Algorithms (CPSC 221)

Vancouver, BC

- Mentored **110+** students during lectures and weekly labs, teaching **data structures and algorithms** in **C++**.

PROJECTS

NBA 3D Scene Reconstruction | Python, PyTorch, OpenCV, RF-DETR, SAM 2, Three.js

Jul 2026 - Present

- Built a **computer vision** pipeline reconstructing monocular NBA clips as interactive 3D scenes in **Three.js**.
- Combined **RF-DETR/SAM 2** detections with **SigLIP** jersey embeddings to track and cluster players by team.
- Used court-keypoint homography and **MotionBERT** 3D pose lifting to reconstruct and animate player meshes.

BizBot (NwHacks 2026) | Supabase, YOLOv8, FastAPI, Next.js, Google Gemini

January 2026

- Won** Best Real-World Ready AI Product from **Block** among **630+** **hackers** by building an autonomous robot photographer with a **YOLOv8** vision loop to auto-trigger photo capture and publish an **AI-filtered** live gallery.

Failure Cloud (Berkeley AI Hackathon 2026) | FastAPI, Next.js, PyBullet, NumPy

May 2026

- Built a natural-language robotics simulator that generates portable URDF/USD failure scenarios for testing robot perception and behavior across platforms; demonstrated with PyBullet RGB-D and point-cloud evaluation.

NBA salary | Spring Boot, JPA, Python, Java, React, TailwindCSS

August 2025 – September 2025

- Built a **Spring Boot** NBA salary API with **JPA/PostgreSQL** for relational data modeling and a React frontend.

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, JavaScript/TypeScript, SQL (PostgreSQL + pgvector)

Frameworks & ML: Spring Boot, React, Node.js, FastAPI, Ray, YOLOv8, OpenCV, RF-DETR, SAM 2, PyTorch

Cloud & Tools: AWS (Lambda/DynamoDB/EC2/Amplify/Cognito/API Gateway), Kafka, Docker, Git, Linux